ECON1050 Tutorial 4

Calculus 1: Limits and Differentiation

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Today's Plan

- Recap: key ideas from Lecture 4
- Tutorial questions: Q1–Q6

Recap: Limits

Core idea

A limit describes the value that $f(x)$ approaches when x approaches a point a.

Notation

$$\lim_{x \longrightarrow a} f(x).$$

The limit depends on values of $f(x)$ close to a, not necessarily on the value of $f(a)$ itself.

Two-sided requirement

When computing a limit, values of x on both sides of a must be considered.

Recap: Univariate Differentiation

Meaning

Differentiation measures the rate of change of one variable in response to a change in another variable. The derivative gives the slope of the tangent line to a function at a point.

Notation

Common notations include

$$f'(x), \qquad y', \qquad \frac{\mathrm{d}y}{\mathrm{d}x}, \qquad \frac{\mathrm{d}f(x)}{\mathrm{d}x}.$$

Newton quotient

For a function $f(x)$, the derivative at a is

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

Recap: Tangent Line, Differentiability and Monotonicity

Tangent line

The tangent line to f at $x = a$ is

$$y - f(a) = f'(a)(x - a).$$

Differentiability

To be differentiable, a function must be continuous. Continuity is necessary, but not sufficient.

Using the first derivative on an interval I

$$\begin{aligned}
&\text{Increasing} &&\iff && f'(x) \geq 0, \quad x \in I,\\
&\text{Strictly increasing} &&\iff && f'(x) > 0, \quad x \in I,\\
&\text{Decreasing} &&\iff && f'(x) \leq 0, \quad x \in I,\\
&\text{Strictly decreasing} &&\iff && f'(x) < 0, \quad x \in I.
\end{aligned}$$

Recap: Rates of Change

Instantaneous rate of change

The derivative gives the instantaneous rate of change:

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

Relative rate of change

The relative rate of change of f at a is

$$\frac{f'(a)}{f(a)}.$$

Recap: Basic Differentiation Rules

Constant and power rules

$$\frac{\mathrm{d}}{\mathrm{d}x} c = 0, \qquad \frac{\mathrm{d}}{\mathrm{d}x} x^a = a x^{a-1}.$$

Constants and sums

$$\frac{\mathrm{d}}{\mathrm{d}x} [cf(x)] = cf'(x), \qquad \frac{\mathrm{d}}{\mathrm{d}x} [f(x) \pm g(x)] = f'(x) \pm g'(x).$$

Recap: Product, Quotient Rules, and Chain Rule

Product rule

If $F(x) = f(x)g(x)$, then

$$F'(x) = f'(x)g(x) + f(x)g'(x).$$

Quotient rule

If $F(x) = \frac{f(x)}{g(x)}$ and $g(x) \neq 0$, then

$$F'(x) = \frac{f'(x)g(x) - g'(x)f(x)}{[g(x)]^2}.$$

Chain rule

Let $y = F(x) = f(g(x))$. Then

$$F'(x) = f'(g(x))g'(x).$$

Recap: Exponential and Logarithmic Derivatives

Exponential functions

$$\frac{\mathrm{d}}{\mathrm{d}x} e^x = e^x, \qquad \frac{\mathrm{d}}{\mathrm{d}x} e^{g(x)} = g'(x) e^{g(x)}.$$

For base $a > 0$,

$$\frac{\mathrm{d}}{\mathrm{d}x} a^x = (\ln a) a^x.$$

Logarithmic functions

$$\frac{\mathrm{d}}{\mathrm{d}x} \ln x = \frac{1}{x}, \qquad \frac{\mathrm{d}}{\mathrm{d}x} \ln(g(x)) = \frac{g'(x)}{g(x)}.$$

Recap: Higher Order Derivatives and Shape

Second derivative

If $f'(x)$ is differentiable, then its derivative is called the second derivative:

$$f''(x) = \frac{\mathrm{d}^2 f(x)}{\mathrm{d}x^2}.$$

Concavity and convexity

On an interval I,

$$\begin{aligned} f''(x) \geq 0 \quad &\Rightarrow \quad f \text{ is convex on } I, \\ f''(x) \leq 0 \quad &\Rightarrow \quad f \text{ is concave on } I. \end{aligned}$$

Tutorial Q1 & Q2

Question 1

Compute the first derivative, i.e. $\dfrac{\mathrm{d}y}{\mathrm{d}x}$, for the following functions:

$$\text{(a)} \quad y = \frac{1}{3}x^9, \qquad \text{(b)} \quad y = 3x^7 + 8, \qquad \text{(c)} \quad y = 5x^3 + 3x - 1.$$

Question 2

Compute the first derivative for the following functions using the product rule or quotient rule:

(a) $y = (x^3 - 1)(x^2 + 2)$

(b) $g(t) = \dfrac{2t + 1}{t^2 + 3}$

(c) $y = 5x^2 + \dfrac{2x}{x^2 + 3}$

(d) $y = (3x - 1)(2x^2 + 3)$

Tutorial Q1 & Q2: Knowledge Used

Knowledge used

- Constant rule: $\dfrac{\mathsf{d}}{\mathsf{d}x}c = 0.$
- Power rule: $\dfrac{\mathsf{d}}{\mathsf{d}x}x^a = ax^{a-1}.$
- Sum rule: differentiate each term separately.

Knowledge used

- Product rule:
$$(fg)' = f'g + fg'.$$
- Quotient rule:
$$\left(\frac{f}{g}\right)' = \frac{f'g - g'f}{g^2}, \qquad g \neq 0.$$
- Sum rule: combine derivative rules term by term.

Tutorial Q1: Solution

Solution

$$
\begin{aligned}
&\text{(a)} \quad y = \frac{1}{3}x^9 \quad \Rightarrow \quad y' = \frac{1}{3}(9)x^{9-1} = 3x^8. \\
&\text{(b)} \quad y = 3x^7 + 8 \quad \Rightarrow \quad y' = 3(7)x^{7-1} + 0 = 21x^6. \\
&\text{(c)} \quad y = 5x^3 + 3x - 1 \quad \Rightarrow \quad y' = 5(3)x^{3-1} + 3(1)x^{1-1} - 0 = 15x^2 + 3.
\end{aligned}
$$

Tutorial Q2: Solution I

Part (a)

Let $f(x) = x^3 - 1$, $g(x) = x^2 + 2$. Then

$$y' = f'(x)g(x) + f(x)g'(x) = 3x^2(x^2 + 2) + (x^3 - 1)(2x).$$

$$= 3x^4 + 6x^2 + 2x^4 - 2x = 5x^4 + 6x^2 - 2x.$$

Part (b)

Let $f(t) = 2t + 1$, $h(t) = t^2 + 3$. Then $f'(t) = 2$ and $h'(t) = 2t$. Using the quotient rule,

$$g'(t) = \frac{f'(t)h(t) - f(t)h'(t)}{[h(t)]^2} = \frac{2(t^2 + 3) - (2t + 1)(2t)}{(t^2 + 3)^2}.$$

$$= \frac{2t^2 + 6 - 4t^2 - 2t}{(t^2 + 3)^2} = \frac{-2t^2 - 2t + 6}{(t^2 + 3)^2}.$$

Tutorial Q2: Solution II

Part (c)

$$y = 5x^2 + \frac{2x}{x^2+3}.$$

$$y' = 10x + \frac{2(x^2+3) - 2x(2x)}{(x^2+3)^2}.$$

$$= 10x + \frac{2x^2 + 6 - 4x^2}{(x^2+3)^2} = 10x + \frac{6 - 2x^2}{(x^2+3)^2}.$$

Part (d)

$$y' = 3(2x^2+3) + (3x-1)(4x) = 6x^2 + 9 + 12x^2 - 4x.$$

$$y' = 18x^2 - 4x + 9.$$

Tutorial Q3

Question 3

Compute the first derivative for the following functions using the chain rule:

$$\text{(a)} \quad y = 10(5 - x^2)^2, \qquad \text{let } u = 5 - x^2,$$

$$\text{(b)} \quad y = (3x^2 + 6x + 6)^{1/2},$$

$$\text{(c)} \quad y = \sqrt{\frac{1}{x} - 1}.$$

Tutorial Q3: Knowledge Used

Knowledge used

- Chain rule for $y = f(g(x))$:
$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}.$$
- Rewrite square roots as fractional powers.
- Apply the power rule to the outer function and differentiate the inner function.

Tutorial Q3: Solution I

Part (a)

Let

$$u(x) = 5 - x^2, \qquad y = 10u^2.$$

Then

$$\frac{\mathrm{d}y}{\mathrm{d}u} = 20u, \qquad \frac{\mathrm{d}u}{\mathrm{d}x} = -2x.$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 20u(-2x) = 20(5 - x^2)(-2x).$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 40x^3 - 200x.$$

Tutorial Q3: Solution II

Part (b)

Let

$$u = 3x^2 + 6x + 6, \qquad y = u^{1/2}.$$

Then

$$\frac{dy}{du} = \frac{1}{2}u^{-1/2}, \qquad \frac{du}{dx} = 6x + 6.$$

$$\frac{dy}{dx} = \frac{1}{2}(3x^2 + 6x + 6)^{-1/2}(6x + 6) = \frac{3x + 3}{(3x^2 + 6x + 6)^{1/2}}.$$

Tutorial Q3: Solution III

Part (c)

$$y = \sqrt{\frac{1}{x} - 1} = \left(\frac{1}{x} - 1\right)^{1/2}.$$

Let

$$u = \frac{1}{x} - 1, \qquad \frac{\mathrm{d}u}{\mathrm{d}x} = -x^{-2}, \qquad y = u^{1/2}.$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{2}u^{-1/2}(-x^{-2}) = -\frac{1}{2}\left(\frac{1}{x} - 1\right)^{-1/2} x^{-2}.$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{2\sqrt{x^3 - x^4}}.$$

Question 4

Compute the second derivatives for the following functions:

$$\text{(a)}\quad z = 120t - \frac{1}{3}t^3, \qquad \text{(b)}\quad y = \sqrt{x}, \qquad \text{(c)}\quad g(t) = \frac{t^2}{t-1}.$$

Tutorial Q4: Knowledge Used

Knowledge used

- First derivative: differentiate the original function once.
- Second derivative: differentiate the first derivative.
- Use the power rule and quotient rule where needed.

Tutorial Q4: Solution I

Part (a)

$$z = 120t - \frac{1}{3}t^3.$$

$$z' = 120 - \frac{1}{3}(3)t^2 = 120 - t^2.$$

$$z'' = -2t.$$

Part (b)

$$y = \sqrt{x} = x^{1/2}.$$

$$y' = \frac{1}{2}x^{-1/2}.$$

$$y'' = -\frac{1}{4}x^{-3/2} = -\frac{1}{4\sqrt{x^3}}.$$

Tutorial Q4: Solution II

Part (c)

$$g(t) = \frac{t^2}{t-1}, \qquad t \neq 1.$$

$$g'(t) = \frac{2t(t-1) - t^2}{(t-1)^2} = \frac{t^2 - 2t}{(t-1)^2}.$$

$$g''(t) = \frac{(2t-2)(t-1)^2 - 2(t-1)(t^2-2t)}{(t-1)^4}.$$

$$= \frac{(2t-2)(t-1) - 2(t^2-2t)}{(t-1)^3} = \frac{2}{(t-1)^3}.$$

Tutorial Q5

Question 5

1. Find the intervals where the following function is strictly increasing:

$$y = (\ln x)^2 - 4.$$

2. For the function

$$f(x) = 2x^3 - 12x^2 + 5,$$

find the intervals where it is increasing and the intervals of concavity/convexity.

Tutorial Q5: Knowledge Used

Knowledge used

- A function is strictly increasing where $f'(x) > 0$.
- A function is increasing where $f'(x) \geq 0$.
- A function is concave where $f''(x) \leq 0$ and convex where $f''(x) \geq 0$.
- Use sign diagrams to identify intervals.

Tutorial Q5: Solution I

Part 1

$$y = (\ln x)^2 - 4, \qquad x > 0.$$

$$y' = 2(\ln x)\frac{1}{x} = \frac{2\ln x}{x}.$$

Since $x > 0$, the sign of y' is determined by $\ln x$:

x	$(0,1)$	1	$(1,\infty)$
$\ln x$	$-$	0	$+$
x	$+$	$+$	$+$
y'	$-$	0	$+$

Therefore,

y is strictly increasing for $x > 1$.

Tutorial Q5: Solution II

Part 2: Increasing intervals

$$f(x) = 2x^3 - 12x^2 + 5.$$

$$f'(x) = 6x^2 - 24x = 6x(x-4).$$

x	$(-\infty, 0)$	0	$(0, 4)$	4	$(4, \infty)$
x	$-$	0	$+$	$+$	$+$
$x-4$	$-$	$-$	$-$	0	$+$
$f'(x)$	$+$	0	$-$	0	$+$

Thus

$f(x)$ is increasing on $(-\infty, 0] \cup [4, \infty)$.

Tutorial Q5: Solution III

Part 2: Concavity and convexity

$$f''(x) = 12x - 24 = 12(x-2).$$

x	$(-\infty, 2)$	2	$(2, \infty)$
$x-2$	$-$	0	$+$
$f''(x)$	$-$	0	$+$

Since $f''(x) \leq 0$ for $x \leq 2$ and $f''(x) \geq 0$ for $x \geq 2$,

$f(x)$ is concave on $(-\infty, 2]$

and convex on $[2, \infty)$.

Tutorial Q6

Question 6

For the function

$$y = \sqrt{x} - x^2,$$

find the equation for the tangent to the graph at the point $x = 4$.

Tutorial Q6: Knowledge Used

Knowledge used

- The tangent slope at $x = a$ is $f'(a)$.
- Tangent line formula:
$$y - f(a) = f'(a)(x - a).$$
- Use the power rule to differentiate $\sqrt{x} = x^{1/2}$.

Tutorial Q6: Solution

Solution

$$f(x) = x^{1/2} - x^2.$$

$$f'(x) = \frac{1}{2}x^{-1/2} - 2x.$$

At $x = 4$,

$$f'(4) = \frac{1}{2}(4)^{-1/2} - 2(4) = \frac{1}{2} \cdot \frac{1}{\sqrt{4}} - 8 = \frac{1}{4} - 8 = -\frac{31}{4}.$$

$$f(4) = \sqrt{4} - 4^2 = 2 - 16 = -14.$$

Using $y - f(4) = f'(4)(x - 4)$:

$$y + 14 = -\frac{31}{4}(x - 4).$$

Therefore,

$$y = -\frac{31}{4}x + 17.$$

Key Takeaways

- The derivative measures instantaneous rate of change and tangent slope.
- Product, quotient and chain rules are essential for differentiating composite expressions.
- The first derivative identifies increasing/decreasing intervals.
- The second derivative identifies concavity and convexity.